

AEROVISION

Real-Time Flight Intelligence & Accessible Airport Portal

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PROBLEM STATEMENT

The Challenges of Modern Air Travel

01 | Fragmented Flight Updates

Passengers struggle with delayed or out-of-sync flight tracking data scattered across different airline systems and boards.

02 | No Road Traffic Predictability

Existing navigation maps show static travel times that fail to account for Bengaluru's severe peak-hour traffic gridlocks on the way to BLR Airport.

03 | Terminal Congestion Blindspots

Travelers face unpredictable security and check-in queue lengths, causing stress and poor time management inside terminal halls.

04 | Inaccessible Assistance Booking

Elderly or disabled passengers endure slow manual paperwork and verification barriers when requesting buggy or wheelchair services.

OUR SOLUTION

AeroVision — A Unified Flight-Driven Intelligence & Navigation App

Live FIDS & Space Radar

An integrated departures/arrivals board coupled with a moving aircraft tracking map that visualizes approaches and departures relative to airport coordinates.

Indoor Pathfinding Navigation

Interactive SVG maps of Terminal 1 and Terminal 2 mapping gates, restaurants, and baggage belts with built-in route planning for passengers.

Flight-Driven Crowd Analytics

Estimates terminal check-in, security, and immigration queue wait-times dynamically based on real-time flight density and passenger load equations.

Assist & Travel Planner

Features browser-based Aadhaar OCR checking to auto-approve senior assistance, and OSRM route calculations using time-of-day traffic multipliers.

TECHNOLOGY STACK

Frontend Stack

- React.js 18 (Component-based architecture)
- Vite (Next-generation frontend toolchain)
- Tailwind CSS (Adaptive design tokens)
- Leaflet.js (Interactive airspace radar)
- Axios (JWT authorization interceptors)
- Tesseract.js (Client-side image OCR)

Backend & DB

- Node.js (Server runtime engine)
- Express.js (REST API endpoint router)
- SQL.js (SQLite-compatible backend DB)
- Socket.IO (WebSocket push event engine)
- node-fetch (External API connection client)
- Helmet & Compression (Security & speed)

External APIs & Feeds

- AeroDataBox API (Schedules & gates)
- OpenWeather API (Airfield METAR weather)
- OpenRouteService (Travel planning geocode)
- OSRM Project (Keyless driving routing)
- Nominatim (Alternative geocoding feed)
- Hugging Face Inference (AI chat completions)

HOW IT WORKS

01 | Flight Search & FIDS

User logs in, enters a flight number. The backend queries AeroDataBox API (or local mock DB) to retrieve schedules, gate numbers, and terminal routes.

02 | Travel Route Plan

The user inputs their location. OSRM computes the driving distance and the backend applies IST-based traffic multipliers for a realistic commute duration.

03 | Queue Wait Estimation

The crowd engine calculates check-in, security, and immigration density levels based on flight counts and passenger loads in a 3-hour window.

04 | Mobility OCR Booking

The user uploads an Aadhaar card. Client-side Tesseract.js extracts the DOB, verifies senior status, and auto-approves wheelchair/buggy booking.

KEY FEATURES

Live Flight Board

Real-time departures and arrivals with dynamic gate, terminal, and status updates, supported by live cron refreshes.

Airspace Radar Map

Interactive map plotting moving flights. Real-time trajectory rendering based on active flight windows (90 mins).

Interactive SVG Terminal

Detailed indoor terminal layout showing gates, lounges, restrooms, and shops with route drawing pathfinding.

Zone Queue Congestion

Real-time queue wait-time calculations for check-in and security driven by passenger volumes and delays.

Traffic Travel Planner

GPS-based route mapping with automatic peak/off-peak travel time adjustments matching Bangalore traffic hours.

AI Chatbot Assistant

Context-aware AI assistant answering passenger queries about gates, airport services, and airport directions.

CHALLENGES & SOLUTIONS

⚠ CHALLENGE

Database Concurrency Crashes

SQL.js wrapper threw errors on `.catch`` calls when executing queries.

Routing API Key Restrictions

TomTom API keys got revoked, blocking geocoding and travel route planning.

Vite CSS Tailwind Purging

Babel JSX compile error broke Hot Module Reloading, causing unstyled components.

Radar Plane Clustering

Simulated planes clustered in static circles right on top of the airport.

Inaccurate Commute Times

Default OSRM route durations ignored Bengaluru's extreme street gridlocks.

✓ SOLUTION

Converted the query method to ``async`` to wrap row outputs in promises, resolving crashes.

Replaced TomTom with OpenRouteService, and integrated keyless OSRM/Nominatim fallbacks.

Fixed raw unicode character syntax in NotificationPanel, cleared Vite cache, and rebuilt CSS.

Created a rolling 90-minute time-window filter for active flights, dispersing planes along vectors.

Implemented dynamic time-of-day traffic multipliers in IST (e.g. 1.95x during peak office hours).

FUTURE ENHANCEMENTS

Web Push Notifications

Integrate the native browser Web Push API to send passengers real-time alerts for gate changes, boarding announcements, and delay notifications directly to their devices.

Camera-Based Queue Vision

Implement computer vision models on terminal security cameras to dynamically count queue lengths and calculate wait times automatically instead of relying on passenger load estimates.

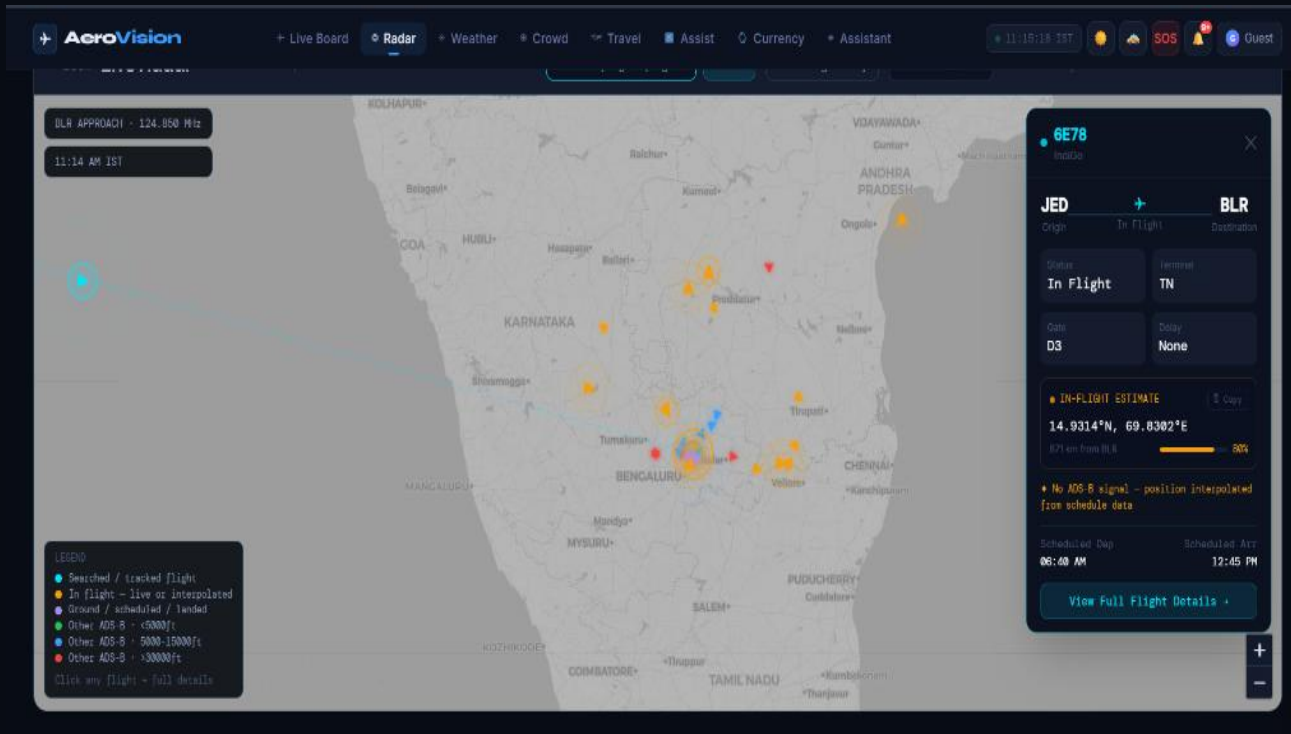
Multi-Airport Scalability

Expand the platform architecture to cover other major domestic hubs in India (like DEL, BOM, and MAA) by dynamically loading specific SVG terminal maps and airport coordinate databases.

Offline Terminal Maps

Cache SVG blueprints and routing tables in a browser service worker (PWA) so passengers can access indoor maps and pathfinding instructions even when offline inside the terminal.

AEROVISION IN ACTION — LIVE RADAR



Live Airspace Radar Console

Direct real-time simulation plots arrivals and departures centered around Bengaluru (VOBL/BLR) airfield airspace.

Active Flight Telemetry

Clickable flight icons open a detailed sidebar showing origin/destination, gates, terminal slots, and speed vectors.

Dynamic Interpolation Engine

Automatically switches to schedule-based route math when direct ADS-B coverage is restricted, ensuring 100% demo uptime.

CONCLUSION

AeroVision

"A production-ready aviation intelligence web application for Kempegowda International Airport (BLR), Bengaluru, integrating live flight boards, terminal pathfinding, passenger crowd density analytics, and smart accessibility booking."

[Accessibility-First Design] · [Verified API Resilience] · [Dynamic Operation Modeling]

Thank You!

Questions & Feedback Welcome